Proposal: The Illusion of Reasoning — Content-Neutral Logic vs. Statistical Pattern Completion

The Seminar/Paper Connection: In my presentation, I highlighted Shanahan's argument that LLMs do not possess true reasoning capabilities. True reasoning (formal logic) is **content-neutral**—the logic holds up regardless of the specific words used. Shanahan posits that LLMs instead rely on **token-level pattern completion** driven by the statistical proximity of words in their training corpus (Distributional Semantics).

The Research Question: Do LLMs actually apply content-neutral logic to solve reasoning tasks, or is their success purely an illusion created by the statistical frequency of the specific words used in the prompt?

The Hypothesis: If Shanahan is correct that LLMs are merely executing pattern completion rather than actual reasoning, the model's accuracy will significantly degrade when standard vocabulary is replaced with pseudo-words, even if the underlying logical structure of the prompt remains completely identical.

Methodology (The Empirical Work):

1. **Dataset Creation:** Build a small dataset of 20–30 standard logical syllogisms or spatial reasoning puzzles. *(Example: "The ball is inside the cup. The cup is inside the box. Is the ball inside the box?")*
2. **The "Scrambler" Dataset:** Create a parallel dataset where the exact same logical structures are used, but the nouns/verbs are replaced with linguistically valid pseudo-words. *(Example: "The florp is inside the blicket. The blicket is inside the wug. Is the florp inside the wug?")*
3. **Prompting:** Run both datasets through an LLM (like GPT-4 or Gemini) using a zero-shot prompt.
4. **Analysis:** Code the responses for accuracy. Compare the failure rates between the two datasets.